T2D Glucose Digital Twin — System Architecture

Page 1: Scope, Multimodal Ingestion Contract, & Hybrid Physiological-Deep Learning Engine

System Boundary & Hypothesis: Investigational engineering prototype combining observable patient context, wearable history, and mechanistic physiological modeling to forecast glucose trajectories and adverse spike risks over a 120-minute horizon. Evaluated on synthetic physiological cohorts (293 subjects); not clinically validated for autonomous medical decisions.

TIER 1 Multimodal Observable Ingestion (Contract S3) & Latent Firewall

Privileged generator internal latents (e.g., secret true insulin kinetics) are strictly quarantined by the latent firewall. The system admits only clinically observable streams:

Static EHR Context
Age, BMI, HbA1c, eGFR, T2D duration, 9 medication class flags (Metformin, SGLT2i, GLP-1 RA, Insulin).

Dynamic CGM Telemetry
5-minute sampled continuous glucose, ROC trend velocity (Δmg/dL/min), sensor flag, warm-up state.

Wearable Biometrics
Continuous Heart Rate (HR), HRV, step counts, sleep stages, and resting HR warm-up baselines.

Dietary & Behavioral Logs
Logged meal carbohydrates (grams), meal timing, absorption slots, and population habit priors.

TIER 2 Hybrid Physiological Prior + Deep Residual Learning

Mechanistic UKF S3 Engine:

Two-compartment Bergman minimal model coupled to a two-stage nonlinear gut absorption ODE. Parameterized by patient-specific EHR priors:

```
dG/dt = -(p_1 + X)G + p_1 G_b + Ra(t) / V_G
dX/dt = -p_2 X + p_3 (I - I_b)
```

Unscented Kalman Filter (UKF) dynamically tracks insulin sensitivity (S_I) and gastric emptying ($t_{\max}$), delivering continuous 120-minute mechanistic forward rollouts.

PR-TFT Deep Residual Architecture:

Physics-Residual Temporal Fusion Transformer (PR-TFT) learns behavioral and physiological residuals over the mechanistic baseline:

```
ŷ_t^{(q)} = ŷ_t^{\text{UKF}} + &rcirc;_t^{(q)}
  (\text{Observable History}, \text{Context})
```

Variable Selection Networks (VSN) filter multimodal inputs; temporal LSTM encodes past dynamics; causal self-attention models long-range multi-horizon dependencies.

Residual Quantile Heads (Multi-Horizon)

Outputs 7 quantiles (**p05, p10, p25, p50, p75, p90, p95**) across 24 discrete 5-minute steps (120 min total). Conformal interval adjustment guarantees strictly valid empirical coverage.

Discrete-Time Spike Hazard Head

Predicts cumulative onset probability $P(\text{spike within } 120\text{ min})$. Calibrated post-training via isotonic regression ($ECE = 0.000094$), driving clinical alert tiers.

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Page 2: Reliable Telemetry Serving, Chunk-Invariant SSE Proxy, & Clinical Workstation

TIER 3 Reliable Inference Serving & Telemetry Proxy Architecture

FastAPI Inference Service (Port 8001)
backend/serve_pr_tft.py serves Blueprint Section 9 JSON contract. Evaluates data-quality gates, runs UKF rollouts + PR-TFT inference, checks quantile monotonicity, and executes stateful AlertMachine with 120-min refractory suppression.

Next.js SSE Proxy (Port 3000)
src/app/api/patients/[id]/stream/route.ts proxies events using createSafeSSETransformStream. Chunk-invariant parsing searches for earliest delimiter (\r\n\r\n, \n\n, \r\r), immune to arbitrary network framing and TCP fragmentation.

Client Hook (useRealtimeTelemetry)
State machine manages connection lifecycles: connecting → connected → reconnecting. Bounded exponential backoff (max 3 retries), envelope schema validation via Zod, and 15-second client staleness detection.

TIER 4 Clinical Workstation Interface

Cohort Directory
/cohort page. Demographic search, status filters (Healthy, Prediabetes, T2D), and quick-pivot patient cards.

Patient Workspace
/patients/[id]. Visualizes replayed synthetic CGM, 24-step forecast ribbon, sensor staleness, and twin parameters.

Risk Panel & Alert Strip
Cumulative adverse event hazard curve with clinical threshold badges (Watch at $p \ge 0.35$, Warning at $p \ge 0.60$).

Model Contributions
Additive decomposition explaining model arithmetic: mechanistic prior, meal absorption, learned residual.

SAFETY BOUNDARY Scenario Exploration (Interactive What-If)

Supported (UI → /api/forecast → /v1/forecasts):
Routes /api/forecast and /v1/forecasts accept hypothetical meal additions/skips and physical activity inputs, re-solving the linearized twin and re-scoring with PR-TFT from the same anchor. The patient workspace shows baseline vs. scenario ribbons, highest median, largest same-time difference and Δprobability.

Clinical Boundary (Enforced):
Every result is labelled “Simulated model response — not a validated causal treatment effect.” Pharmacological scenarios return SCENARIO_UNSUPPORTED (422). Counterfactual treatment effects have not been validated in humans and must not guide therapy.

Fail-Closed Safety Architecture: If the neural network fails (OOM, timeout), the service falls back to the mechanistic UKF twin-only trajectory (fallback_used: "twin_only") with monotonic quantiles and suppresses hazard alerts. If CGM coverage is insufficient (<6h), forecast and risk are strictly nulled.

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Page 3: Patient-Specific Hybrid Synergies, Calibrated Uncertainty, & Scientific Limitations

CORE DIFFERENTIATION — Why a Physics-Residual Hybrid Architecture?

- **Mass Conservation & Physiological Grounding:** Pure black-box neural networks frequently hallucinate impossible glucose spikes or cliff-drops. The UKF minimal model guarantees mass conservation and physiological bounds.
- **Patient-Specific Personalization:** UKF continuously tracks individual metabolic traits (EHR priors S_I insulin sensitivity and $t_{\max}$ absorption kinetics) from longitudinal history.
- **Deep Residual Learning:** The PR-TFT transformer does not need to re-learn basic human physiology; it focuses entirely on modeling complex residual behaviors (delayed dawn phenomenon, stress, meal composition).
- **Interpretability for Clinicians:** Decomposes predictions into observable physiological drivers and learned corrections without black-box opacity.

EVIDENCE — Validation Benchmark & Calibrated Uncertainty

- **Evaluated Validation Partition:** 44 validation patients (47,629 decision anchors) from pre-registered split_293.json.
- **Trajectory Error (RMSE mg/dL):**
  - **30-min:** 12.36 (MAE 8.43, MARD 7.22%)
  - **60-min:** 19.53 (MAE 12.43, MARD 10.16%)
  - **120-min:** 25.03 (MAE 16.76, MARD 13.66%)
- **Conformal Uncertainty:** 80% empirical interval coverage is **80.00%** (mean width 37.3 mg/dL).
- **Hazard Calibration:** Isotonic-calibrated spike hazard achieves AUROC **0.9226**, AUPRC **0.6314**, and Expected Calibration Error (ECE) **0.000094**.

SCIENTIFIC INTEGRITY — Negative Validation Gates & Explicit Limitations

1. Gate 3 Realism Audit: FAILED
Historical AUROC **0.9320** / Duration-corrected AUROC **0.9290** vs. threshold **≤ 0.65**. Discriminator easily separates synthetic ODE series from mock-real CGM (top tell: jerk_sd). Factorial drift controls prove generator AR(1) drift explains part of curvature differences, but not the full multivariate failure. No fabricated smoothing applied.

2. Baseline Audit: UNVERIFIED
Demo baseline files (Persistence RMSE120 50.66, UKF 40.20) evaluated an 18-patient test subset under an earlier demo run. Because populations are not aligned, direct superiority claims on the 44-subject validation split are marked cohort-unverified.

3. EW-01 Protocol: UNEXECUTED
Formal comparative early warning evaluation remains governance-blocked pending co-author sign-offs. The 44-patient validation split was used for model tuning and calibration; it does not represent independent human generalization.

Investigational Research Prototype: This submission establishes verifiable engineering excellence (42 Jest suites, 413 tests; 134 backend tests). All negative findings are disclosed in full accordance with scientific peer-review standards.